

**Memo 2: Isothermal Reactor Design with Python 3D Sensitivity Visualization
Covering Chapter 4 – Part 1****Due Date: 16/03/2026****OVERVIEW**

Design an **isothermal fixed-bed reactor** to produce your assigned product at the target production rate. Develop both a computational model (POLYMATH) and a 3D visualization (Python) showing how reactor volume and conversion vary with temperature. This memo demonstrates your ability to design reactors and communicate complex multi-parameter relationships through visualization.

Memo 2 has two main sections:

SECTION 1: POLYMATH ISOTHERMAL REACTOR MODEL

SECTION 2: PYTHON 3D SENSITIVITY VISUALIZATION

Grading:

- POLYMATH model and analysis: 50 points
- Python 3D sensitivity visualization: 30 points
- **Documentation & presentation:** 20 points
- **Total: 100 points**

PART 1: POLYMATH ISOTHERMAL REACTOR MODEL

Develop an isothermal reactor model to estimate the reactor volume (or catalyst weight) for the production of your product using your proposed feedstock at your prescribed production rate.

Model Specifications:

- **Reactor Type:** Packed bed (fixed-bed) reactor, modeled as plug flow (PFR)
- **Basis:** Your stoichiometric and kinetic data from Memo 1
- **Operation:** Isothermal (constant temperature)
- **Assumptions:**
 - Single reaction (byproducts added in Memo 4)
 - Negligible pressure drop (added in Memo 3)
 - Steady-state operation
 - Ideal gas behavior (or liquid if applicable)

Design Calculation

For **five representative temperatures** (within your typical operating range):

For each temperature:

1. **Solve POLYMATH ODE** to generate conversion vs. W profile
2. **Determine catalyst weight (W)** required for your target conversion
3. **Calculate reactor volume**
4. **Create a single plot** for all temperatures:
 - X-axis: Catalyst weight (kg)
 - Y-axis: Conversion (%)
 - Mark the point where your target conversion is achieved
5. **Document your design selection:**
 - Which temperature did you choose? Why?
 - What is your final reactor volume?
 - How sensitive is this to temperature changes?

PART 2: PYTHON 3D SENSITIVITY VISUALIZATION

Create a 3D surface plot showing conversion as a function of BOTH temperature and reactor volume (catalyst weight). This visualization reveals the multi-parameter design space and helps you understand trade-offs between:

- Higher temperature → faster reaction → smaller reactor (lower capital cost)
- Lower temperature → slower reaction → larger reactor (higher capital cost)

This is how engineers visualize complex design spaces in practice.

In your memo, include written analysis of the 3D surface:

Answer these questions:

1. What does the surface shape tell you?
 - Where is conversion most sensitive to temperature?
 - Where is conversion most sensitive to catalyst weight?
 - Are there regions of diminishing returns?
2. How do the two parameters compete?
 - Can you achieve the same conversion with:
 - High T, low W?
 - Low T, high W?
 - What are the trade-offs (cost, selectivity, etc.)?
3. Where is your design point?
 - Locate your selected operating point on the surface
 - Why did you choose this point?
 - What happens if you move slightly in temperature or weight?
4. What would you change and why?
 - Is your design robust (small changes don't cause problems)?
 - Are you operating on a steep part of the surface (sensitive) or flat (robust)?
 - Could you achieve your target conversion with a smaller reactor at slightly higher T?

Submit your 3D visualization as:

- High-resolution image file (.png, .jpg, or .pdf)
- Embedded in your memo PDF
- Large enough to see details clearly (suggest 6"×6" or larger)
- Include figure caption with product name, temperature range, weight range

Also submit:

- Python source code (as appendix or supplementary file)
- Data table used to generate surface (.csv or .xlsx)
- Any notes on plotting methodology

PART 3: OPTIONAL ASPEN PLUS SIMULATION

Simulate your reactor using the Aspen Plus_RPLUG reactor block:

- Use the conversion determined in Part 1
- Model the isothermal reactor with your kinetics
- Plot the effect of conversion on reactor heat duty
- Include results and discussion
- This provides validation of your POLYMATH model

DELIVERABLE SUMMARY

Submit by March 26, 2025 at 10:00 AM:

1. **Single PDF file** containing:
 - Full memo with POLYMATH analysis
 - High-resolution 3D surface plot
 - Interpretation and discussion
 - ASPEN PLUS output and discussion
2. **Supplementary files** (as attachments):
 - POLYMATH model (.npt file)
 - Python source code (.py file)
 - Data table (.csv file)

Late submission: 50% credit only